Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Frank Chaves and Sons Dairy #1

Physical address of dairy:

Number and Street	City	County	Zip Code
13080 E Harney LN	Lodi	San Joaquin	95240

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/05/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X063-X260-XX14-XXXX

B. OPERATORS

Chaves, Laurinda

Operator name: Chaves, Laurinda

Telephone no.: (209) 368-1740 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12754 E Harney LN	Lodi	CA	95240

This operator is responsible for paying permit fees.

C. OWNERS

Chaves, Laurinda

Legal owner name: Chaves, Laurinda

Telephone no.: (209) 368-1740 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12754 E Harney LN	Lodi	CA	95240

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	140	150	30	0
Number under roof	470	90	0	0	0	0
Maximum number	470	90	140	150	30	0
Average number	470	90	140	150	30	0
Avg live weight (lbs)	1,300	1,400	1,000	500		

Predominant milk cow breed: Holstein

Average milk production: 64 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 15,909.29 *tons per reporting period*

Total nitrogen from manure: 198,780.47 *lbs per reporting period*

After ammonia losses (30% loss applied): 139,146.33 *lbs per reporting period*

Total phosphorus from manure: 32,823.84 *lbs per reporting period*

Total potassium from manure: 88,156.60 *lbs per reporting period*

Total salt from manure: 241,995.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 3,588,750 *gallons*

Total nitrogen generated: 16,178.09 *lbs*

Total phosphorus generated: 2,266.48 *lbs*

Total potassium generated: 29,234.62 *lbs*

Total salt generated: 138,471.09 *lbs*

	3,588,750	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	3,588,750	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
C1D1	Ground water
C1D2	Ground water
C1D3	Ground water
C1I1	Ground water
C1I2	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/15/2023	Corral solids	260.00 *ton*	Dry-weight	10.1		22,400.00	6,190.00	35,800.00		22.30

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	10,471.55	2,893.70	16,735.78	104,248.04
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	10,471.55	2,893.70	16,735.78	104,248.04

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
C1	40	40	2	both	X063-X260-XX14-XXXX
C2	24	24	2	both	X063-X260-XX14-XXXX
C4	21	21	2	both	X063-X240-XX10-XXXX X063-X240-XX30-XXXX
C5	15	15	2	both	X063-X240-XX39-XXXX X063-X240-XX40-XXXX X063-X240-XX41-XXXX
C6	3	3	1	both	X063-X240-XX39-XXXX X063-X240-XX40-XXXX
Totals for areas that were used for application	103	103	9		
Totals for areas that were not used for application					
Land application area totals	103	103	9		

B. CROPS AND HARVESTS

C1

Field name: C1

10/27/2022: Winter Forage

Crop: Winter Forage Acres planted: 40 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	677.00 *ton*	Dry-weight		67.4	16,700.00	4,200.00	63,000.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	163.50	40.50	300.00	0.00
Total actual harvest content	16.93	184.29	46.35	695.21	1,335.25

C1

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	1,080.00 *ton*	Dry-weight		64.6	14,700.00	2,130.00	15,200.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	270.40	39.00	275.60	0.00
Total actual harvest content	27.00	281.01	40.72	290.56	1,261.66

C2

Field name: C2

10/27/2022: Winter Forage

Crop: Winter Forage Acres planted: 24 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	383.00 *ton*	Dry-weight		78.0	20,700.00	5,400.00	83,000.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	136.50	36.00	270.00	0.00
Total actual harvest content	15.96	145.35	37.92	582.80	849.62

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	600.00 *ton*	Dry-weight		71.8	14,300.00	2,440.00	19,700.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	210.60	36.40	286.00	0.00
Total actual harvest content	25.00	201.63	34.40	277.77	1,128.00

C4

Field name: C4

10/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 21 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	356.00 *ton*	Dry-weight		70.2	25,300.00	4,600.00	61,000.00		13.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	226.50	40.50	273.00	0.00
Total actual harvest content	16.95	255.62	46.48	616.32	1,404.40

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	546.00 *ton*	Dry-weight		69.0	11,300.00	1,150.00	19,300.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	182.00	39.00	309.40	0.00
Total actual harvest content	26.00	182.16	18.54	311.12	1,047.80

C5

Field name: C5

10/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 15 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	248.30 *ton*	Dry-weight		70.2	25,300.00	4,600.00	61,000.00		13.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	226.50	40.50	273.00	0.00
Total actual harvest content	16.55	249.60	45.38	601.81	1,371.34

C5

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	390.00 *ton*	Dry-weight		69.0	11,300.00	1,150.00	19,300.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	26.00	182.00	39.00	309.40	0.00
Total actual harvest content	26.00	182.16	18.54	311.12	1,047.80

C6

Field name: C6

10/28/2022: Winter Forage

Crop: Winter Forage Acres planted: 3 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	49.70 *ton*	Dry-weight		70.2	25,300.00	4,600.00	61,000.00		13.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	226.50	40.50	273.00	0.00
Total actual harvest content	16.57	249.81	45.42	602.30	1,372.45

NUTRIENT BUDGET

A. LAND APPLICATIONS

C1 - 10/27/2022: Winter Forage

Field name: C1

Crop: Winter Forage — Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Separator solids	143.31	36.52	225.99	1,336.66	260.00 *ton*
Application event totals		143.31	36.52	225.99	1,336.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	55.54	8.79	135.19	657.92	540,000.00 *gal*
Application event totals		55.54	8.79	135.19	657.92	

C1 - 05/26/2023: Corn, silage

Field name: C1

Crop: Corn, silage — Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.45	0.00	0.00	0.00	2,403,000.00 *gal*
C1I2	Ground water	0.79	0.00	0.00	0.00	2,403,000.00 *gal*
Application event totals		2.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	178.76	51.25	120.54	0.00	400.00 *ton*
Application event totals		178.76	51.25	120.54	0.00	

C1 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.36	0.00	0.00	0.00	2,241,000.00 *gal*
C1I2	Ground water	0.74	0.00	0.00	0.00	2,241,000.00 *gal*
Application event totals		2.09	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	64.02	8.37	94.97	439.36	540,000.00 *gal*
C1I1	Ground water	1.36	0.00	0.00	0.00	2,241,000.00 *gal*
C1I2	Ground water	0.74	0.00	0.00	0.00	2,241,000.00 *gal*
Application event totals		66.12	8.37	94.97	439.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.36	0.00	0.00	0.00	2,241,000.00 *gal*
C1I2	Ground water	0.74	0.00	0.00	0.00	2,241,000.00 *gal*
Application event totals		2.09	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	64.02	8.37	94.97	439.36	540,000.00 *gal*
C1I1	Ground water	1.36	0.00	0.00	0.00	2,241,000.00 *gal*
C1I2	Ground water	0.74	0.00	0.00	0.00	2,241,000.00 *gal*
Application event totals		66.12	8.37	94.97	439.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.36	0.00	0.00	0.00	2,241,000.00 *gal*
C1I2	Ground water	0.74	0.00	0.00	0.00	2,241,000.00 *gal*
Application event totals		2.09	0.00	0.00	0.00	

C2 - 10/27/2022: Winter Forage

Field name: C2

Crop: Winter Forage — Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Separator solids	143.31	36.52	225.99	1,336.66	156.00 *ton*
Application event totals		143.31	36.52	225.99	1,336.66	

02/16/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	46.28	7.32	112.66	548.27	270,000.00 *gal*
Application event totals		46.28	7.32	112.66	548.27	

C2 - 06/12/2023: Corn, silage

Field name: C2

Crop: Corn, silage — Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	148.97	42.71	100.45	0.00	200.00 *ton*
Application event totals		148.97	42.71	100.45	0.00	

05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.63	0.00	0.00	0.00	1,620,000.00 *gal*
C1I2	Ground water	0.89	0.00	0.00	0.00	1,620,000.00 *gal*
Application event totals		2.52	0.00	0.00	0.00	

C2 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.46	0.00	0.00	0.00	1,448,000.00 *gal*
C1I2	Ground water	0.80	0.00	0.00	0.00	1,448,000.00 *gal*
Application event totals		2.26	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	48.91	6.39	72.55	335.63	247,500.00 *gal*
C1I1	Ground water	1.47	0.00	0.00	0.00	1,458,000.00 *gal*
C1I2	Ground water	0.80	0.00	0.00	0.00	1,458,000.00 *gal*
Application event totals		51.18	6.39	72.55	335.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.31	0.00	0.00	0.00	1,296,000.00 *gal*
C1I2	Ground water	0.71	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		2.02	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	48.91	6.39	72.55	335.63	247,500.00 *gal*
C1I1	Ground water	1.31	0.00	0.00	0.00	1,296,000.00 *gal*
C1I2	Ground water	0.71	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		50.92	6.39	72.55	335.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.31	0.00	0.00	0.00	1,296,000.00 *gal*
C1I2	Ground water	0.71	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		2.02	0.00	0.00	0.00	

C4 - 10/28/2022: Winter Forage

Field name: C4

Crop: Winter Forage

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Separator solids	157.49	40.13	248.34	1,468.86	150.00 *ton*
Application event totals		157.49	40.13	248.34	1,468.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	52.90	8.37	128.75	626.59	270,000.00 *gal*
Application event totals		52.90	8.37	128.75	626.59	

C4 - 06/06/2023: Corn, silage

Field name: C4

Crop: Corn, silage

Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	143.01	41.00	96.43	0.00	168.00 *ton*
Application event totals		143.01	41.00	96.43	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.68	0.00	0.00	0.00	1,458,000.00 *gal*
C1I2	Ground water	0.92	0.00	0.00	0.00	1,458,000.00 *gal*
Application event totals		2.60	0.00	0.00	0.00	

C4 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	45.73	5.98	67.84	313.83	202,500.00 *gal*
C1I1	Ground water	1.49	0.00	0.00	0.00	1,296,000.00 *gal*
C1I2	Ground water	0.81	0.00	0.00	0.00	1,296,000.00 *gal*
Application event totals		48.04	5.98	67.84	313.83	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.24	0.00	0.00	0.00	1,080,000.00 *gal*
C1I2	Ground water	0.68	0.00	0.00	0.00	1,080,000.00 *gal*
Application event totals		1.92	0.00	0.00	0.00	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	45.73	5.98	67.84	313.83	202,500.00 *gal*
C1I1	Ground water	1.24	0.00	0.00	0.00	1,080,000.00 *gal*
C1I2	Ground water	0.68	0.00	0.00	0.00	1,080,000.00 *gal*
Application event totals		47.65	5.98	67.84	313.83	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.24	0.00	0.00	0.00	1,080,000.00 *gal*
C1I2	Ground water	0.68	0.00	0.00	0.00	1,080,000.00 *gal*
Application event totals		1.92	0.00	0.00	0.00	

09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.24	0.00	0.00	0.00	1,080,000.00 *gal*
C1I2	Ground water	0.68	0.00	0.00	0.00	1,080,000.00 *gal*
Application event totals		1.92	0.00	0.00	0.00	

C5 - 10/28/2022: Winter Forage

Field name: C5

Crop: Winter Forage — Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Separator solids	143.31	36.52	225.99	1,336.66	97.50 *ton*
Application event totals		143.31	36.52	225.99	1,336.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/18/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	61.71	9.76	150.21	731.02	225,000.00 *gal*
Application event totals		61.71	9.76	150.21	731.02	

C5 - 06/05/2023: Corn, silage

Field name: C5

Crop: Corn, silage — Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	143.01	41.00	96.43	0.00	120.00 *ton*
Application event totals		143.01	41.00	96.43	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.92	0.00	0.00	0.00	1,188,000.00 *gal*
C1I2	Ground water	1.04	0.00	0.00	0.00	1,188,000.00 *gal*
Application event totals		2.96	0.00	0.00	0.00	

C5 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	42.68	5.58	63.31	292.91	135,000.00 *gal*
C1I1	Ground water	1.61	0.00	0.00	0.00	999,000.00 *gal*
C1I2	Ground water	0.88	0.00	0.00	0.00	999,000.00 *gal*
Application event totals		45.17	5.58	63.31	292.91	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.31	0.00	0.00	0.00	810,000.00 *gal*
C1I2	Ground water	0.71	0.00	0.00	0.00	810,000.00 *gal*
Application event totals		2.02	0.00	0.00	0.00	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	42.68	5.58	63.31	292.91	135,000.00 *gal*
C1I1	Ground water	1.31	0.00	0.00	0.00	810,000.00 *gal*
C1I2	Ground water	0.71	0.00	0.00	0.00	810,000.00 *gal*
Application event totals		44.70	5.58	63.31	292.91	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.31	0.00	0.00	0.00	810,000.00 *gal*
C1I2	Ground water	0.71	0.00	0.00	0.00	810,000.00 *gal*
Application event totals		2.02	0.00	0.00	0.00	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
C1I1	Ground water	1.31	0.00	0.00	0.00	810,000.00 *gal*
C1I2	Ground water	0.71	0.00	0.00	0.00	810,000.00 *gal*
Application event totals		2.02	0.00	0.00	0.00	

C6 - 10/28/2022: Winter Forage

Field name: C6

Crop: Winter Forage

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Separator solids	143.31	36.52	225.99	1,336.66	19.50 *ton*
Application event totals		143.31	36.52	225.99	1,336.66	

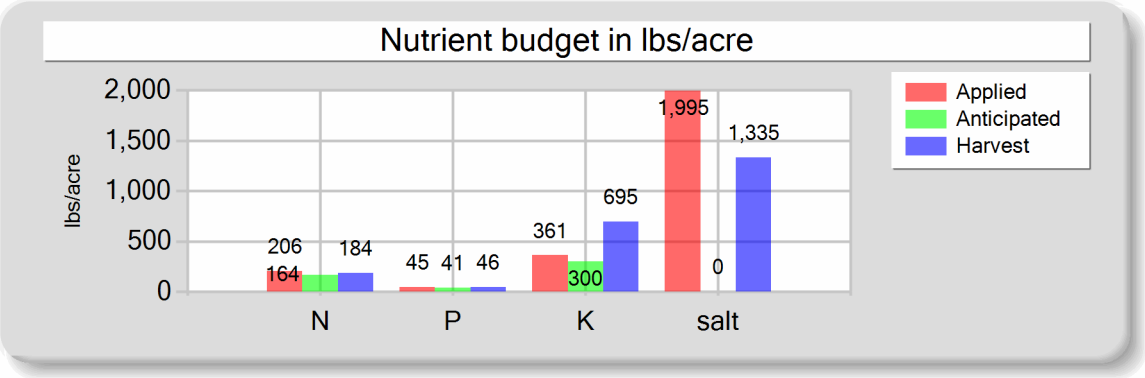
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/18/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	46.28	7.32	112.66	548.27	33,750.00 *gal*
Application event totals		46.28	7.32	112.66	548.27	

B. NUTRIENT BUDGET

C1 - 10/27/2022: Winter Forage

Field name: C1 Crop: Winter Forage Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.31	36.52	225.99	1,336.66
Process wastewater	55.54	8.79	135.19	657.92
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.85	45.30	361.18	1,994.58
Anticipated crop nutrient removal	163.50	40.50	300.00	0.00
Actual crop nutrient removal	184.29	46.35	695.21	1,335.25
Nutrient balance	21.57	-1.04	-334.03	659.33
Applied to removed ratio	1.12	0.98	0.52	1.49

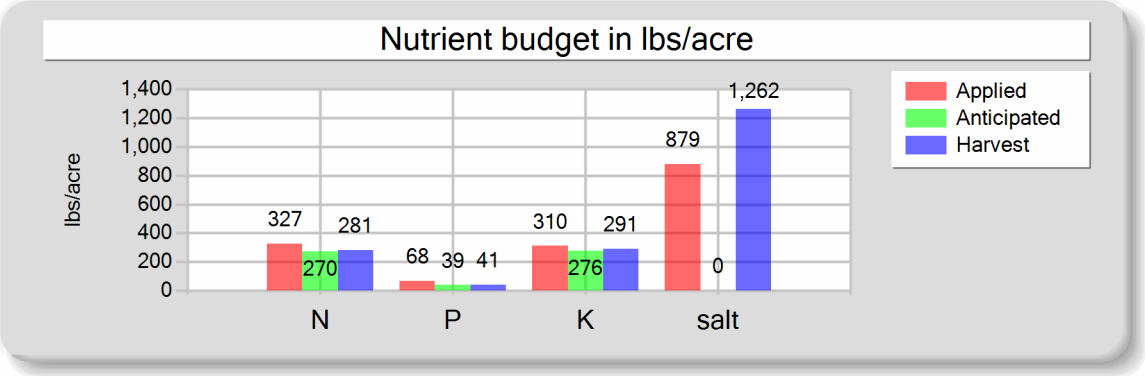
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
540,000.00	*gallons*
19.89	*acre-inches*
0.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

C1 - 05/26/2023: Corn, silage

Field name: C1 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.76	51.25	120.54	0.00
Process wastewater	128.04	16.74	189.94	878.73
Fresh water	12.72	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	326.52	67.99	310.48	878.73
Anticipated crop nutrient removal	270.40	39.00	275.60	0.00
Actual crop nutrient removal	281.01	40.72	290.56	1,261.66
Nutrient balance	45.52	27.27	19.92	-382.93
Applied to removed ratio	1.16	1.67	1.07	0.70

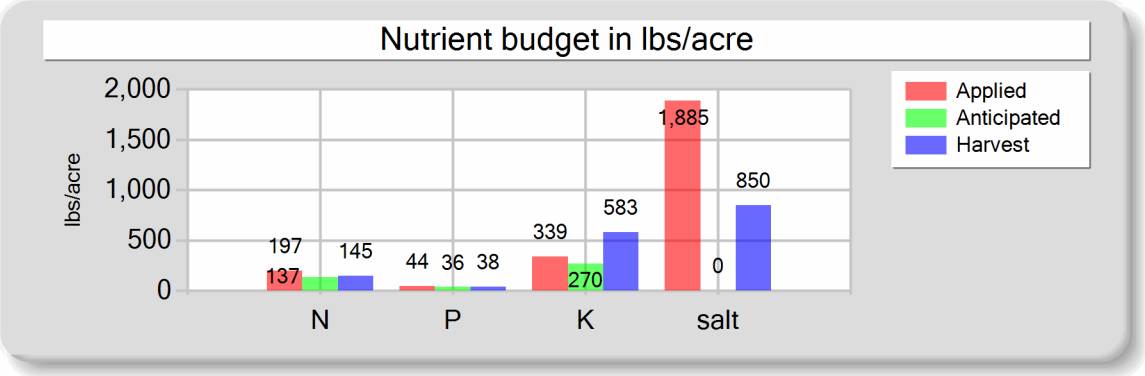
Fresh water applied	
27,216,000.00	*gallons*
1,002.27	*acre-inches*
25.06	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
0.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

C2 - 10/27/2022: Winter Forage

Field name: C2 Crop: Winter Forage Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.31	36.52	225.99	1,336.66
Process wastewater	46.28	7.32	112.66	548.27
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.60	43.84	338.65	1,884.93
Anticipated crop nutrient removal	136.50	36.00	270.00	0.00
Actual crop nutrient removal	145.35	37.92	582.80	849.62
Nutrient balance	51.25	5.92	-244.15	1,035.30
Applied to removed ratio	1.35	1.16	0.58	2.22

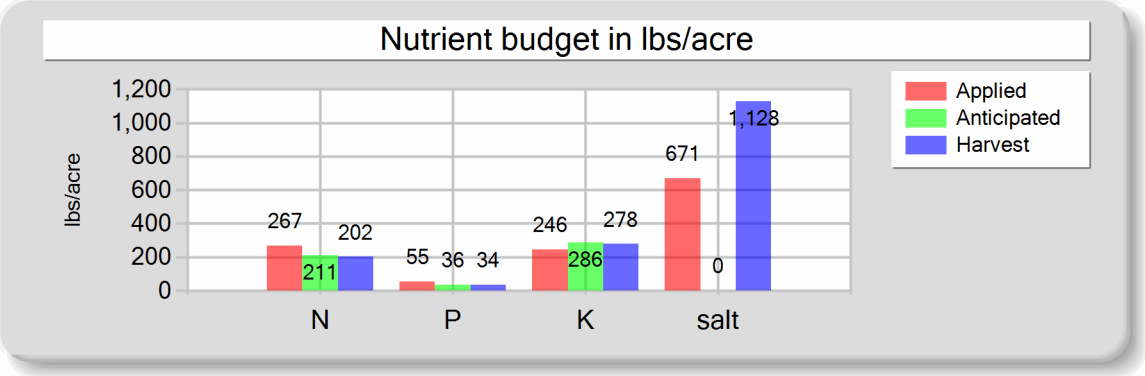
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
270,000.00 *gallons*
9.94 *acre-inches*
0.41 *inches/acre*

Total harvests for the crop
1 *harvests*

C2 - 06/12/2023: Corn, silage

Field name: C2 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	148.97	42.71	100.45	0.00
Process wastewater	97.81	12.79	145.09	671.25
Fresh water	13.11	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	266.89	55.50	245.54	671.25
Anticipated crop nutrient removal	210.60	36.40	286.00	0.00
Actual crop nutrient removal	201.63	34.40	277.77	1,128.00
Nutrient balance	65.26	21.09	-32.23	-456.75
Applied to removed ratio	1.32	1.61	0.88	0.60

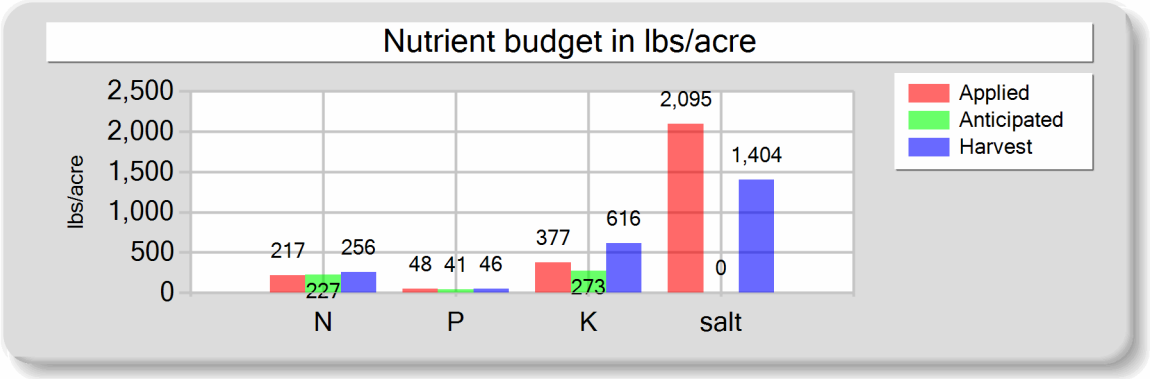
Fresh water applied	
16,828,000.00	*gallons*
619.72	*acre-inches*
25.82	*inches/acre*

Process wastewater applied	
495,000.00	*gallons*
18.23	*acre-inches*
0.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

C4 - 10/28/2022: Winter Forage

Field name: C4 Crop: Winter Forage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	157.49	40.13	248.34	1,468.86
Process wastewater	52.90	8.37	128.75	626.59
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	217.38	48.50	377.09	2,095.45
Anticipated crop nutrient removal	226.50	40.50	273.00	0.00
Actual crop nutrient removal	255.62	46.48	616.32	1,404.40
Nutrient balance	-38.24	2.02	-239.23	691.04
Applied to removed ratio	0.85	1.04	0.61	1.49

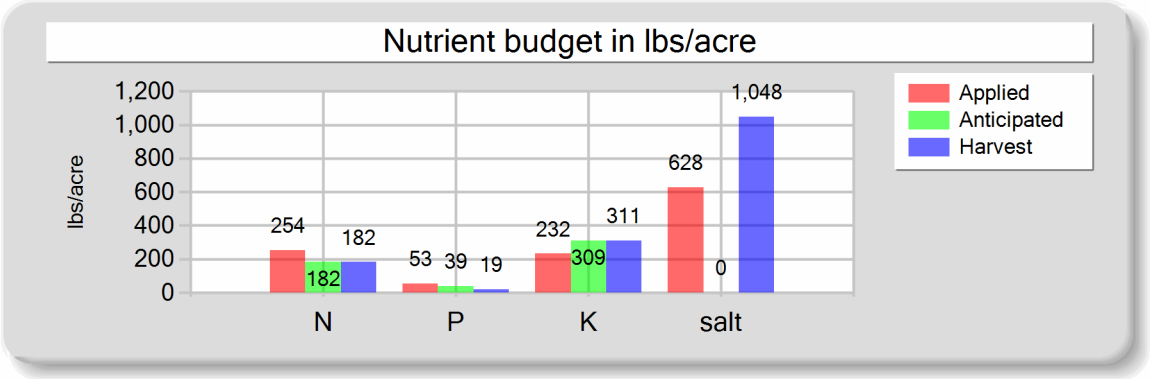
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
270,000.00 *gallons*
9.94 *acre-inches*
0.47 *inches/acre*

Total harvests for the crop
1 *harvests*

C4 - 06/06/2023: Corn, silage

Field name: C4 Crop: Corn, silage Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.01	41.00	96.43	0.00
Process wastewater	91.46	11.96	135.67	627.66
Fresh water	12.59	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	254.06	52.96	232.10	627.66
Anticipated crop nutrient removal	182.00	39.00	309.40	0.00
Actual crop nutrient removal	182.16	18.54	311.12	1,047.80
Nutrient balance	71.91	34.42	-79.01	-420.14
Applied to removed ratio	1.39	2.86	0.75	0.60

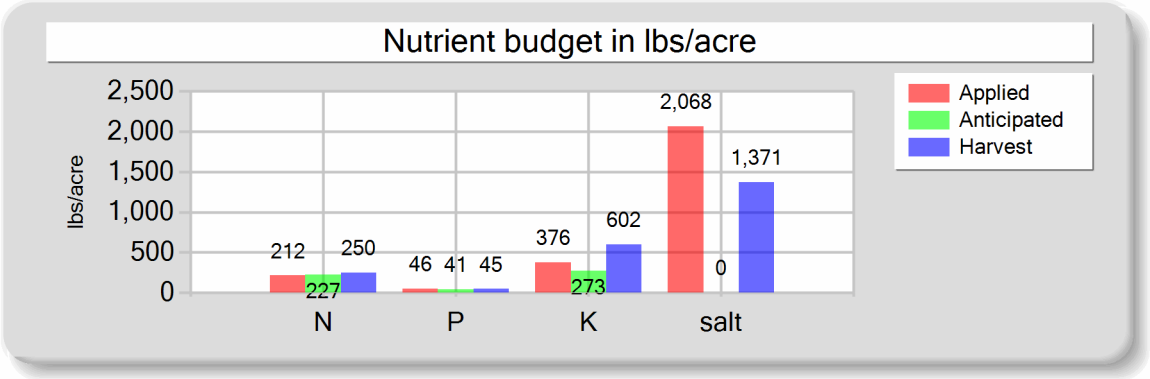
Fresh water applied	
14,148,000.00	*gallons*
521.02	*acre-inches*
24.81	*inches/acre*

Process wastewater applied	
405,000.00	*gallons*
14.91	*acre-inches*
0.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

C5 - 10/28/2022: Winter Forage

Field name: C5 Crop: Winter Forage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.31	36.52	225.99	1,336.66
Process wastewater	61.71	9.76	150.21	731.02
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	212.02	46.28	376.20	2,067.68
Anticipated crop nutrient removal	226.50	40.50	273.00	0.00
Actual crop nutrient removal	249.60	45.38	601.81	1,371.34
Nutrient balance	-37.58	0.90	-225.61	696.34
Applied to removed ratio	0.85	1.02	0.63	1.51

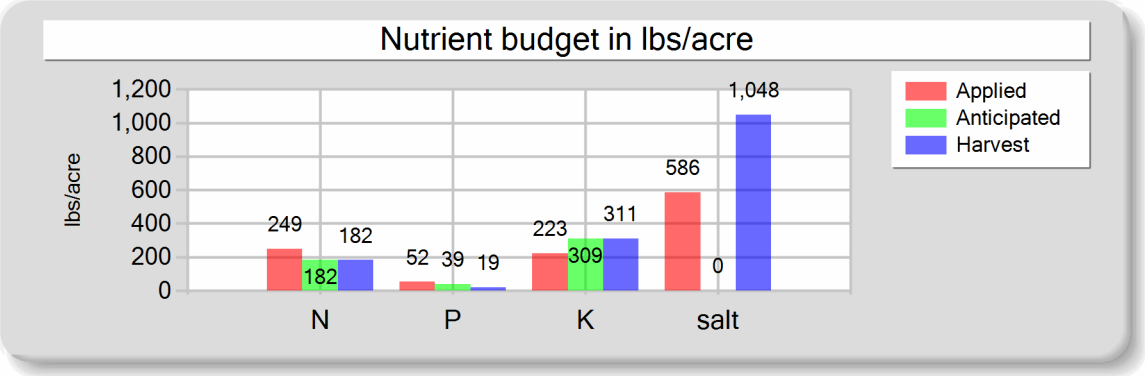
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
225,000.00 *gallons*
8.29 *acre-inches*
0.55 *inches/acre*

Total harvests for the crop
1 *harvests*

C5 - 06/05/2023: Corn, silage

Field name: C5 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.01	41.00	96.43	0.00
Process wastewater	85.36	11.16	126.63	585.82
Fresh water	13.53	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	248.90	52.16	223.06	585.82
Anticipated crop nutrient removal	182.00	39.00	309.40	0.00
Actual crop nutrient removal	182.16	18.54	311.12	1,047.80
Nutrient balance	66.74	33.62	-88.06	-461.98
Applied to removed ratio	1.37	2.81	0.72	0.56

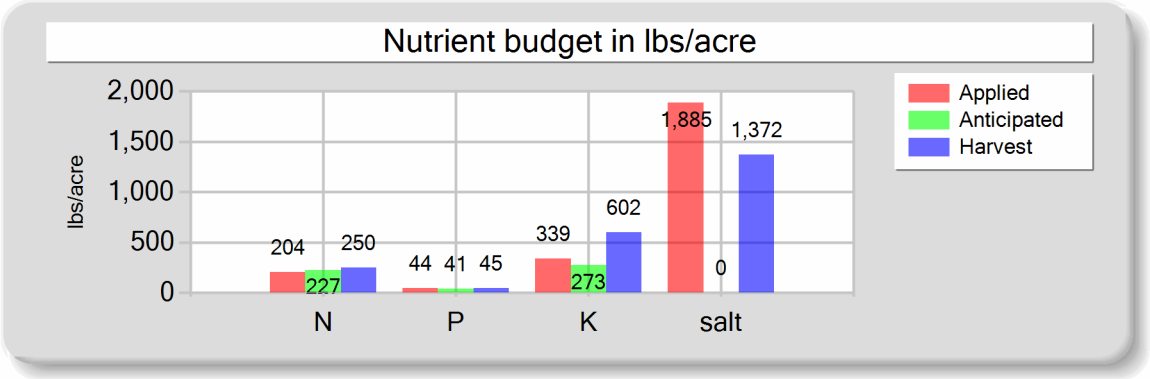
Fresh water applied	
10,854,000.00	*gallons*
399.72	*acre-inches*
26.65	*inches/acre*

Process wastewater applied	
270,000.00	*gallons*
9.94	*acre-inches*
0.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

C6 - 10/28/2022: Winter Forage

Field name: C6 Crop: Winter Forage Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.31	36.52	225.99	1,336.66
Process wastewater	46.28	7.32	112.66	548.27
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	203.60	43.84	338.65	1,884.93
Anticipated crop nutrient removal	226.50	40.50	273.00	0.00
Actual crop nutrient removal	249.81	45.42	602.30	1,372.45
Nutrient balance	-46.21	-1.58	-263.65	512.48
Applied to removed ratio	0.82	0.97	0.56	1.37

Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
33,750.00	*gallons*
1.24	*acre-inches*
0.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2313026A

Sample and source description: T2313026A

Sample date: 11/09/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,800.00	5,300.00	32,800.00	9,800.00	5,190.00	2,930.00	4,100.00	727.00		19.40
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3132035D

Sample and source description: T3132035D

Sample date: 05/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,800.00	6,250.00	14,700.00							0.00
DL	1.00	1.00	1.00							1.00

T3265029B

Sample and source description: T3265029B

Sample date: 09/22/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 10.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	6,190.00	35,800.00	10,900.00	6,500.00	1,810.00	3,730.00	724.00		22.30
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3053010A

Sample and source description: T3053010A

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	493.00	308.00	0.00	0.00	78.00	1,200.00								7,720.00	5,840
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3132035E

Sample and source description: T3132035E

Sample date: 05/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	563.00	378.00	0.00	0.00	75.60	666.00								5,880.00	4,080
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3265029A

Sample and source description: T3265029A

Sample date: 09/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	568.00	319.00	0.00	0.29	74.30	843.00	123.00	84.90	81.70	2,440.00	0.00	220.00	339.00	5,790.00	3,900
DL	1.00	1.00	1.00	0.10	1.00	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

C. FRESH WATER ANALYSES

C1D1

C1D1

T3314014A

Sample description: T3314014A

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.16	0.00	4.16								396.00	0
DL	1.00	0.50	1.00								100.00	1

C1D2

T3314014B

Sample description: T3314014B

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.12	0.00	1.12								232.00	0
DL	1.00	0.50	1.00								100.00	1

C1D3

T3314014C

Sample description: T3314014C

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.97	0.00	1.97	16.80	7.58	25.10	88.00	0.00	9.99	23.90	270.00	150
DL	1.00	0.50	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

C1I1

C1I1

T3314014D

Sample description: T3314014D

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.90	0.00	2.90								311.00	0
DL	1.00	0.50	1.00								100.00	1

C1I2

T3314014E

Sample description: T3314014E

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.58	0.00	1.58								248.00	0
DL	1.00	0.50	1.00								100.00	1

D. SOIL ANALYSES

C1

T3354017A

Sample and source description: T3354017A

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			123.00		0.00	0.00	
DL			1.00		1.00	1.00	

C2

T3354017B

Sample and source description: T3354017B

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			128.00		0.00	0.00	
DL			1.00		1.00	1.00	

C4

T3354017B

Sample and source description: T3354017B

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			128.00		0.00	0.00	
DL			1.00		1.00	1.00	

T3354017C

Sample and source description: T3354017C

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			126.00		0.00	0.00	
DL			1.00		1.00	1.00	

C5

C5

T3354017C

Sample and source description: T3354017C

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			126.00		0.00	0.00	
DL			1.00		1.00	1.00	

C6

T3354017C

Sample and source description: T3354017C

Sample date: 12/20/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			126.00		0.00	0.00	
DL			1.00		1.00	1.00	

E. PLANT TISSUE ANALYSES

C1 - 10/27/2022: Winter Forage

T3132035B

Sample and source description: T3132035B

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,700.00	4,200.00	63,000.00		12.10
DL	1.00	1.00	1.00		1.00

C1 - 05/26/2023: Corn, silage

C1 - 05/26/2023: Corn, silage

T3265029C

Sample and source description: T3265029C

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	2,130.00	15,200.00		6.60
DL	1.00	1.00	1.00		1.00

C2 - 10/27/2022: Winter Forage

T3132035A

Sample and source description: T3132035A

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,700.00	5,400.00	83,000.00		12.10
DL	1.00	1.00	1.00		1.00

C2 - 06/12/2023: Corn, silage

T3265029D

Sample and source description: T3265029D

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	2,440.00	19,700.00		8.00
DL	1.00	1.00	1.00		1.00

C4 - 10/28/2022: Winter Forage

T3132035C

Sample and source description: T3132035C

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,300.00	4,600.00	61,000.00		13.90
DL	1.00	1.00	1.00		1.00

C4 - 06/06/2023: Corn, silage

T3291032A

Sample and source description: T3291032A

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	1,150.00	19,300.00		6.50
DL	1.00	1.00	1.00		1.00

C5 - 10/28/2022: Winter Forage

T3132035C

Sample and source description: T3132035C

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,300.00	4,600.00	61,000.00		13.90
DL	1.00	1.00	1.00		1.00

C5 - 06/05/2023: Corn, silage

T3291032A

Sample and source description: T3291032A

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	1,150.00	19,300.00		6.50
DL	1.00	1.00	1.00		1.00

C6 - 10/28/2022: Winter Forage

T3132035C

Sample and source description: T3132035C

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,300.00	4,600.00	61,000.00		13.90
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

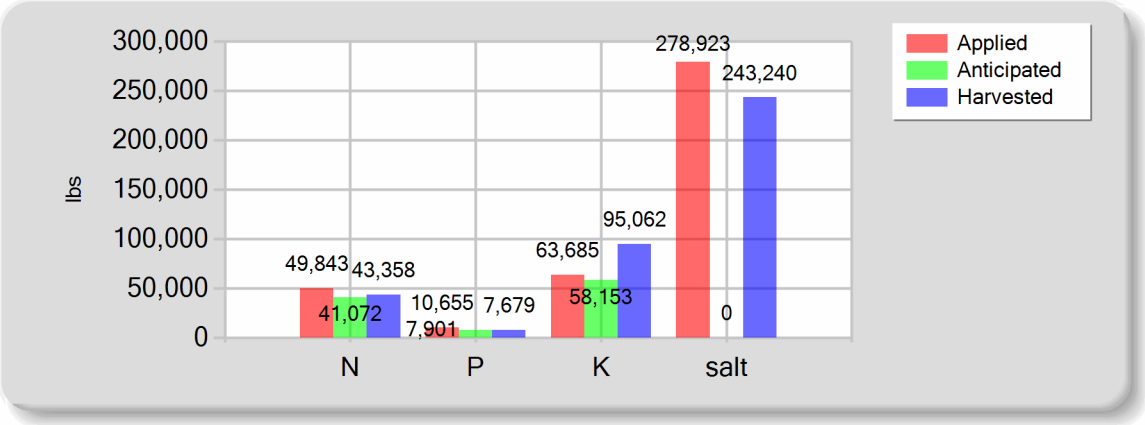
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

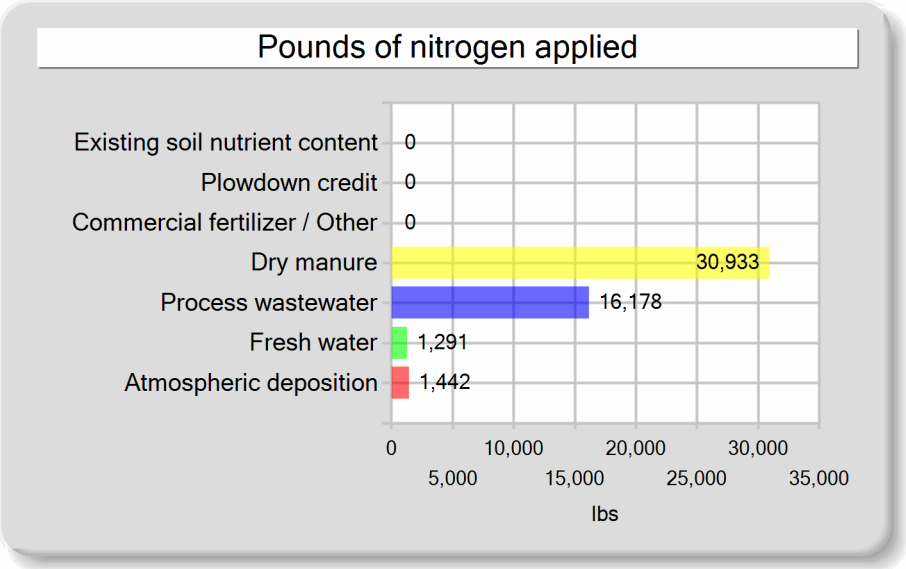
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

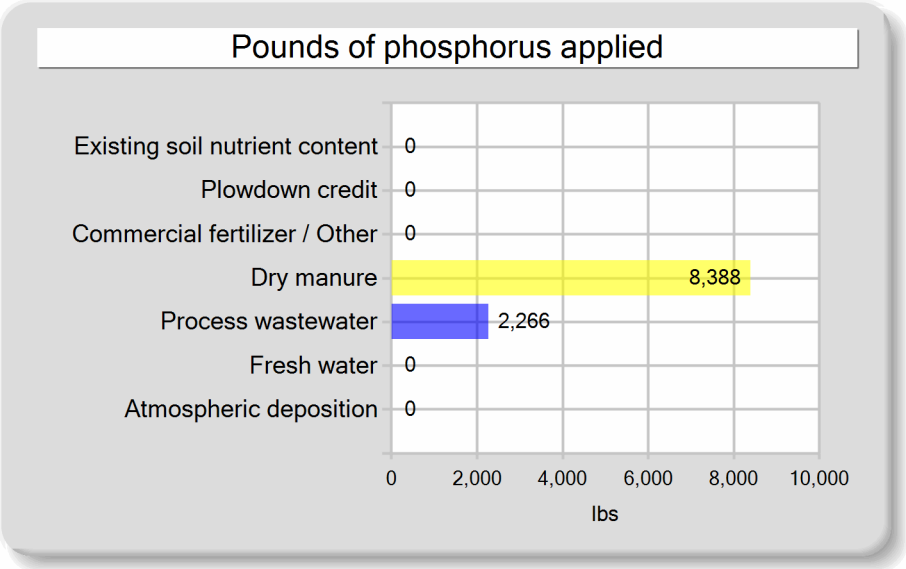
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	30,932.67	8,388.09	34,450.50	140,452.12
Process wastewater	16,178.09	2,266.48	29,234.62	138,471.09
Fresh water	1,290.66	0.00	0.00	0.00
Atmospheric deposition	1,442.00	0.00	0.00	0.00
Total nutrients applied	49,843.42	10,654.58	63,685.12	278,923.21
Anticipated crop nutrient removal	41,071.90	7,901.10	58,153.40	0.00
Actual crop nutrient removal	43,358.29	7,678.66	95,061.62	243,239.82
Nutrient balance	6,485.13	2,975.92	-31,376.50	35,683.39
Applied to removed ratio	1.15	1.39	0.67	1.15

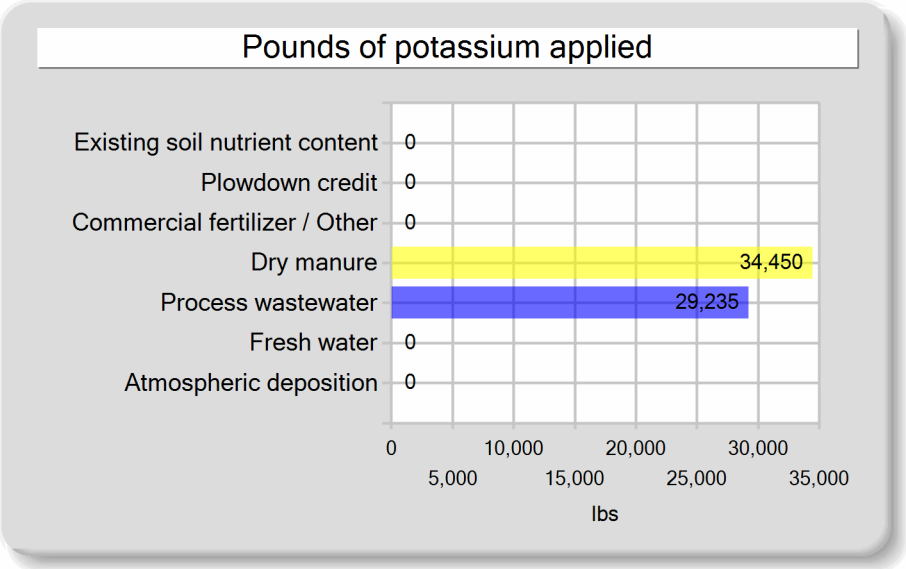
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

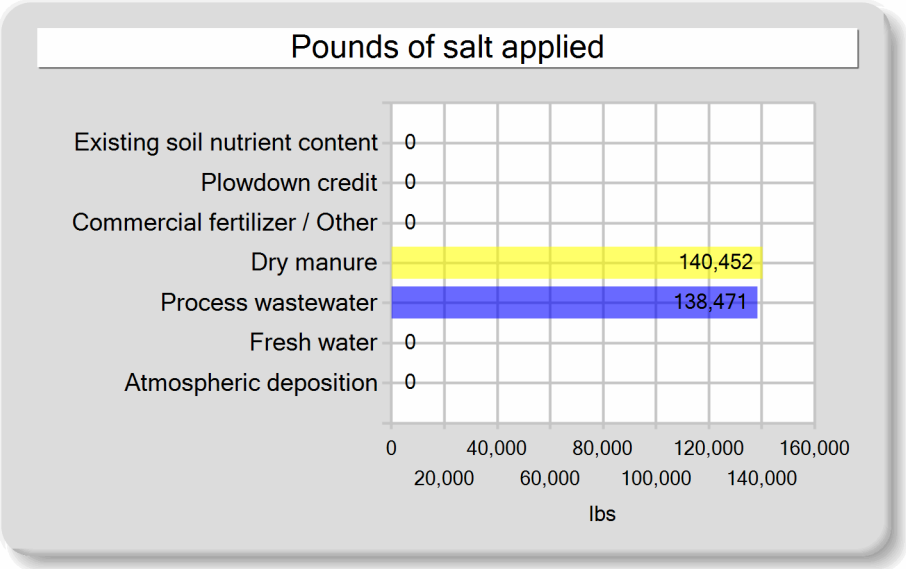


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Laurinda S. Chaves
SIGNATURE OF OWNER OF FACILITY

Laurinda Chaves
PRINT OR TYPE NAME

6/19/24
DATE

Laurinda S. Chaves
SIGNATURE OF OPERATOR OF FACILITY

SAME AS OWNER
PRINT OR TYPE NAME

6/19/24
DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ___STANLEY CHAVES_____

Name of Dairy Facility: _FRANK CHAVES DAIRY #1_____

Facility Address: __13080 E. HARNEY LANE_____LODI_____95240__
Number and Street City Zip Code

Contact Person Name and Phone Number:_STANLEY CHAVES _____(209) 329-4243__
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: __ALBERT CUNHA_____

Address of Hauling Company /Person:_1327 W. HARNEY LANE_____LODI_____95240___
Number and Street City Zip Code

Contact Person:_____ALBERT CUNHA_____(209) 334-1851_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_FRANK CHAVES DAIRY #2__12021 E. HARNEY LANE_LODI__95240__(209) 329-4243____
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____SAME_____
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: __10/15 – 11/15/24_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 2.24 % P 0.619% K 3.58 %

Manure: 260_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): _22.30%_____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: Hauler's Estimate

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Stanley Chaves Date: 2/21/24

Hauler's Signature: Albert Bueno Date: 2/21/24

DENELE ANALYTICAL, INC.
865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: JCRK, INC.
(209)327-0992
Sampled By: Stanley Chaves
Project ID:

Client: FRANK CHAVES & SONS DAIRY #1
Mailing address: 12754 E. HARNEY LANE
(not facility address) LODI, CA 95240
Phone#: (209) 329-4243

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/9/23	1:00p	C1D1	Dom. Well	11.6°c	20 min.	X				
2	✓	1:30p	C1D2	✓	12.6°	✓	X				
3	✓	2:00p	C1D3	✓	11.0°	✓					H14
4	✓	2:30p	C1I1	Ag Well	13.6°	✓	X				
5	✓	3:00p	C1I2	✓	11.6°	✓	X				
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Jacquie Captein	JCRK, INC.		11/10/23 8:00a		
2nd		DENELE ANALYTICAL, INC.	11-10-23 1:00			T3314014

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
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www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Frank Chaves & Sons Dairy #1
12754 E. Harney Ln.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/13/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
C1D1	T3314014A	Electrical Conductivity (EC)	EPA 120.1 and	0.396	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.16	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	253	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4			
Electrical Conductivity (EC)	93.8			1.53
Nitrate Nitrogen (NO3-N)	100	106 (T3314001A)	103 (T3314001A)	3.44

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ELAP Certificate No.: 2714

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12754 E. Harney Ln.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/13/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
C1D2	T3314014B	Electrical Conductivity (EC)	EPA 120.1 and	0.232	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.12	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	148	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4			
Electrical Conductivity (EC)	93.8			1.53
Nitrate Nitrogen (NO3-N)	100	106 (T3314001A)	103 (T3314001A)	3.44

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12754 E. Harney Ln.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/13/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
C1D3	T3314014C	Electrical Conductivity (EC)	EPA 120.1 and	0.270	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	88.0	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	9.99	mg/L
		Sodium (Na)	EPA 200.7	25.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	7.58	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.97	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	150	mg/L
		Calcium (Ca)	EPA 200.7	16.8	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	23.9	mg/L
		Soluble Salts (SALT-SOL)	Calculation	172	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	94.4			0.000 (T3314013A)	0.000 (null)	
Calcium (Ca)	105	94.5 (T3314002A)	94.4 (T3314002A)			0.102
Chloride (Cl)	101	91.1 (T3314001A)	90.2 (T3314001A)			0.927
Magnesium (Mg)	94.1	103 (T3314002A)	102 (T3314002A)			1.14
Nitrate Nitrogen (NO3-N)	100	106 (T3314001A)	103 (T3314001A)			3.44
Sodium (Na)	93.2	110 (T3314002A)	107 (T3314002A)			2.17
Sulfate (SO4)	108	126 (T3314001A)	126 (T3314001A)			0.615
Total Dissolved Solids (TDS)	102			50.0 (T3313017B)	50.0 (Dup)	0.000

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12754 E. Harney Ln.
Lodi, CA 95240

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/13/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 13.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
C1I1	T3314014D	Electrical Conductivity (EC)	EPA 120.1 and	0.311	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.90	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	199	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4			
Electrical Conductivity (EC)	93.8			1.53
Nitrate Nitrogen (NO3-N)	100	106 (T3314001A)	103 (T3314001A)	3.44

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SUBMITTED BY: Jacquie
ANALYZED DATE: 11/13/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
C1I2	T3314014E	Electrical Conductivity (EC)	EPA 120.1 and	0.248	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.58	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	159	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.4			
Electrical Conductivity (EC)	93.8			1.53
Nitrate Nitrogen (NO3-N)	100	106 (T3314001A)	103 (T3314001A)	3.44

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